

Antoun Adel Shoukry
2200724

Assignment "2" Mechanics

1) $V_E = 86 \text{ cm/s}$

$$\vec{V}_D = \vec{V}_{DB} + \vec{V}_B$$

$$V_D \angle -135^\circ = 150 \omega_{BD} \angle -165 + 90^\circ + 45 \omega_{AB} \angle 120 - 90^\circ$$

Real Parts:

$$V_D \cos(135) = 150 \omega_{BD} \cos(-75) + 45 \omega_{AB} \cos(30)$$

Imag Part:

$$V_D \sin(-135) = 150 \omega_{BD} \sin(-75) + 45 \omega_{AB} \sin(30)$$

$$\omega_{BD} = 0.09 \omega_{AB} \quad \omega_{AB} = -9.37 \text{ rad/s}$$

$$\therefore V_D = 470.6 \text{ cm/s}$$

$$\sum \text{Power} = 0$$

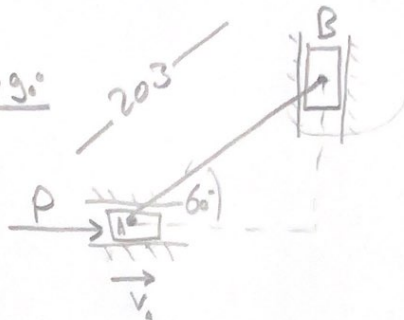
$$10^3 \times 4.7 \cos 0 + 1.5 \times 10^3 \times 0.86 \cos(180) + T_{AB}(9.37) = 0$$

$$T_{AB} = -3629 \text{ N.m} = 3629 \text{ opp to } \omega_{AB}$$

2) $m_A = 4.812 \text{ kg}$, $m_B = 2.72 \text{ kg}$, $\mu = 0.08$, $v_A = 91.5 \text{ cm/s}$, $a_A = 152.5 \text{ cm/s}^2$

$$\vec{V}_A = \vec{V}_{AB} + \vec{V}_B$$

$$91.5 \angle 0^\circ = 20.3 \omega_{AB} \angle -120 + 90^\circ + V_B \angle 90^\circ$$



Real Parts:

$$91.5 = 17.58 \omega_{AB}, \omega_{AB} = 5.2 \text{ C.C.W}$$

• Imag Parts:

$$0 = 20.3(5.2)(-0.5) + V_B, \vec{V}_B = 52.78 \text{ m/s} \uparrow$$

$$\vec{A}_A = \vec{A}_{AB} + \vec{A}_B$$

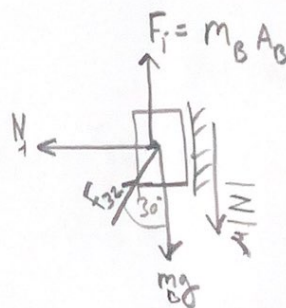
$$152.5 \angle 0 = (5.2)^2 (20.3) \angle -120^\circ + \alpha_{AB} (20.3) \angle -120 + 90^\circ + A_B \angle 90^\circ$$

$$\alpha_{AB} = -6.9 \text{ rad/s}^2, \vec{A}_B = -547.5 \text{ cm/s}^2 \downarrow$$

Link AB: $F_{43} = F_{23}$

Block B:

$$F_x \rightarrow -\frac{1}{2} F_{32} + N_1 = 0 \quad (1)$$



$$F_y \rightarrow 2.72(0.5475) + \frac{\sqrt{3}}{2} F_{32} - 0.05|N_1| - 2.72(9.81) = 0 \quad (2)$$

$$N_1 = 7.1 \text{ N}$$

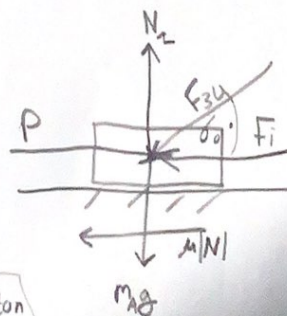
$$F_{32} = 14.3 \text{ N}$$

Block A:

$$F_y \rightarrow N_2 - 4.812(9.8) - \frac{\sqrt{3}}{2}(14.3) = 0 \rightarrow N_2 = 30.2 \text{ N}$$

$$F_x \rightarrow P - m_A a_A - \mu(30.2) - \frac{1}{2}(14.3) = 0$$

$$\therefore P = 42.3 \text{ Newton}$$



$$\begin{array}{c} \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \end{array}$$

The diagram shows a central pulley with a radius of r . Three strings are attached to the pulley:

- A horizontal string on the left, fixed to a wall at point E , with tension T_1 .
- A vertical string on the right, fixed to a wall at point F , with tension T_2 .
- A diagonal string extending downwards and to the left, fixed to a wall at point G , with tension T_3 .

 The pulley is free to rotate, as indicated by the curved arrow. The strings are labeled S_1 , S_2 , and S_3 at their ends. The pulley's radius is labeled r at the bottom.

$$\vec{V}_{ED} = \vec{V}_{RA} + \vec{V}_{HR}$$

Real Part:

Ing Part.

$$\vec{r}_{ED} = \vec{r}_{RA} + \vec{r}_{HR}$$

$$A_{EA} = 3036 \text{ cm/s}^2, A_{ED} = -946 \text{ cm/s}^2$$

Force analysis (separation method):

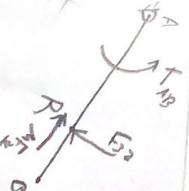
$$G_{-28} F_{23} - N - N' = 0$$

$$-0.82F_z - 0.15\dot{N} - 0.15|N| = 20$$

$$F_{T3} = 244 \text{ N}, \quad N_1 = 16 \text{ N}, \quad N_2 = 16.1 \text{ N}$$

$$\sum_{A \rightarrow T} \chi_A - 24h(S_4)(2) = 0$$

$$T_{AB} = 2.6 \text{ N}$$



4) $N_A = 200 \text{ rpm}$...

$$\vec{R}_1 + \vec{R}_2 + \vec{R}_3 = 0$$

$$10.2 \angle 30^\circ + R_2 \angle \theta_2 + 15.2 \angle 90^\circ = 0$$

$$\text{Real Parts: } 10.2 \cos 30^\circ + R_2 \cos \theta_2 + 0 = 0 \quad (1)$$

Imag Parts:

$$10.2 \sin 30^\circ + R_2 \sin \theta_2 + 15.2 = 0 \quad (2)$$

$$\therefore R_2 = 22.1 \text{ cm}, \theta_2 = -113.5^\circ = 246.5^\circ$$

$$\vec{V}_A + \vec{V}_C = 0$$

$$10.2 \omega_A \angle 30^\circ + 22.1 \omega_C \angle -114^\circ + V_S \angle -114^\circ = 0$$

Real Parts:

$$10.2 \left(2\pi \times \frac{200}{60} \right) \cos(-60^\circ) + 22.1 \omega_C \cos(-24^\circ) + V_S \cos(-114^\circ) = 0$$

Imag Parts:

$$10.2 \left(2\pi \times \frac{200}{60} \right) \sin(-60^\circ) + 22.1 \omega_C \sin(-24^\circ) + V_S \sin(-114^\circ) = 0$$

$$\omega_C = -7.8 \text{ rad/sec}, V_S = -127.4 \text{ cm/sec}$$

$$\vec{A}_A + \vec{A}_C = 0$$

$$A_A^r + A_C^r + A_C^t + A_C^r + A_C^t + A_C^r = 0$$

$$-10.2 \omega_A^2 \angle 30^\circ - 22.1 \omega_C^2 \angle -114^\circ + 22.1 \alpha_C \angle -114^\circ + 2 V_S \omega_C \angle -114^\circ + a_S \angle -114^\circ = 0$$

Real Parts:

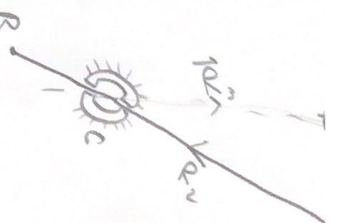
$$20.2 \alpha_C - 0.4 a_S = 1512.3 \quad (1)$$

Imag Parts:

$$-0.3 \alpha_C - 0.3 a_S = 1817.2 \quad (2)$$

$$\rightarrow \alpha_C = 29.1 \text{ rad/s}^2$$

$$a_S = -2310 \text{ cm/s}^2$$



$OA = 10.2 \text{ cm}$
 $AC = 22.1 \text{ cm}$
 $OC = 15.2 \text{ cm}$

me 4)

$$\vec{A}_c = 3420 \angle -153.4^\circ$$

$$\sum P_{\text{Power}} = 0:$$

$$T \times \frac{200}{60} \times 2\pi + 6.35 \times \frac{3420}{100} \sqrt{6} \cos(\theta_c - \theta_f) + \frac{6.35 \times 305^2}{12} \angle -23.1^\circ = 0$$

$$T = 6.21 \text{ N.m}$$

$$5) \cdot \vec{V}_{CH} + \vec{V}_{HA} = \vec{V}_c$$

$$\omega_{AB} = 3 \text{ rad/s}, \alpha_{AB} = 5 \text{ rad/s}^2$$

$$V_s \angle -30^\circ + 0.75 \times 3 \angle -30 + 90^\circ = 0.5 \omega_{CD} \angle 30 + 90^\circ$$

Real Parts:

$$V_s \cos(-30^\circ) + \frac{9}{4} \cos 60^\circ = 0.5 \omega_{CD} \cos 180^\circ \quad (1)$$

Imag Parts:

$$V_s \sin(-30^\circ) + \frac{9}{4} \sin 60^\circ = 0.5 \omega_{CD} \sin 180^\circ \quad (2)$$

$$V_s = 3.897 \text{ m/s} \rightarrow$$

$$\omega_{DC} = -9 \text{ rad/s} \text{ CW}$$

$$\vec{A}_{CH} + \vec{A}_{HA} = \vec{A}_c$$

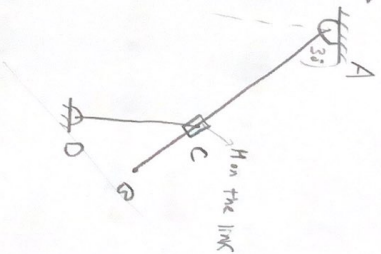
$$A_s \angle -30^\circ + 2 \times 3 \times 3.897 \angle -30 + 90^\circ - 0.75 \times 3^2 \angle -30 + 10.75 \angle -30 + 90^\circ = -0.5 \angle -90^\circ \times \frac{200}{60} \times \frac{12}{12} \quad (3)$$

Real Parts:

$$A_s \cos(-30^\circ) + 23.82 \cos 60^\circ - 6.75 \cos(-30^\circ) + 3.75 \cos 60^\circ = -40.5 \cos 90^\circ + 0.5 \times 60 \cos 180^\circ \quad (4)$$

$$A_s \sin(-30^\circ) + 23.82 \sin 60^\circ - 6.75 \sin(-30^\circ) + 3.75 \sin 60^\circ = -40.5 \sin 90^\circ + 0.5 \times 60 \sin 180^\circ \quad (5)$$

$$(4) \rightarrow A_s = 134.7 \text{ m/s}^2, \alpha_{CD} = -248.82 \text{ rad/s}^2$$



$$6). \vec{V}_A = \vec{V}_{AB} + \vec{V}_B$$

$$V_A \angle -30^\circ = 20 \omega_{AB} \angle 150^\circ + 50 \angle 0^\circ$$

Real Parts:

$$0 = 20 \omega_{AB} \cos(240^\circ) + 50 \rightarrow \omega_{AB} = -5 \text{ rad/sec}$$

Imag Parts:

$$-V_A = 20 \omega_{AB} \sin(240^\circ) \rightarrow V_A = 86.6 \text{ cm/s} \downarrow$$

$$\vec{A}_A = \vec{A}_{AB} + \vec{A}_B$$

$$A_A \angle -30^\circ = -20 \alpha_{AB} \angle 150^\circ + 20 \alpha_{AB} \angle 150^\circ + 30 \angle 0^\circ$$

Real Parts:

$$0 = -500 \cos(150^\circ) + 20 \alpha_{AB} \cos(240^\circ) + 30 \rightarrow \alpha_{AB} = 46.3 \text{ rad/s}^2$$

Imag Parts:

$$-A_A = -500 \sin(150^\circ) + 20 (46.3) \sin(240^\circ) + 0 \rightarrow A_A = 1051.9 \text{ cm/s}^2$$

